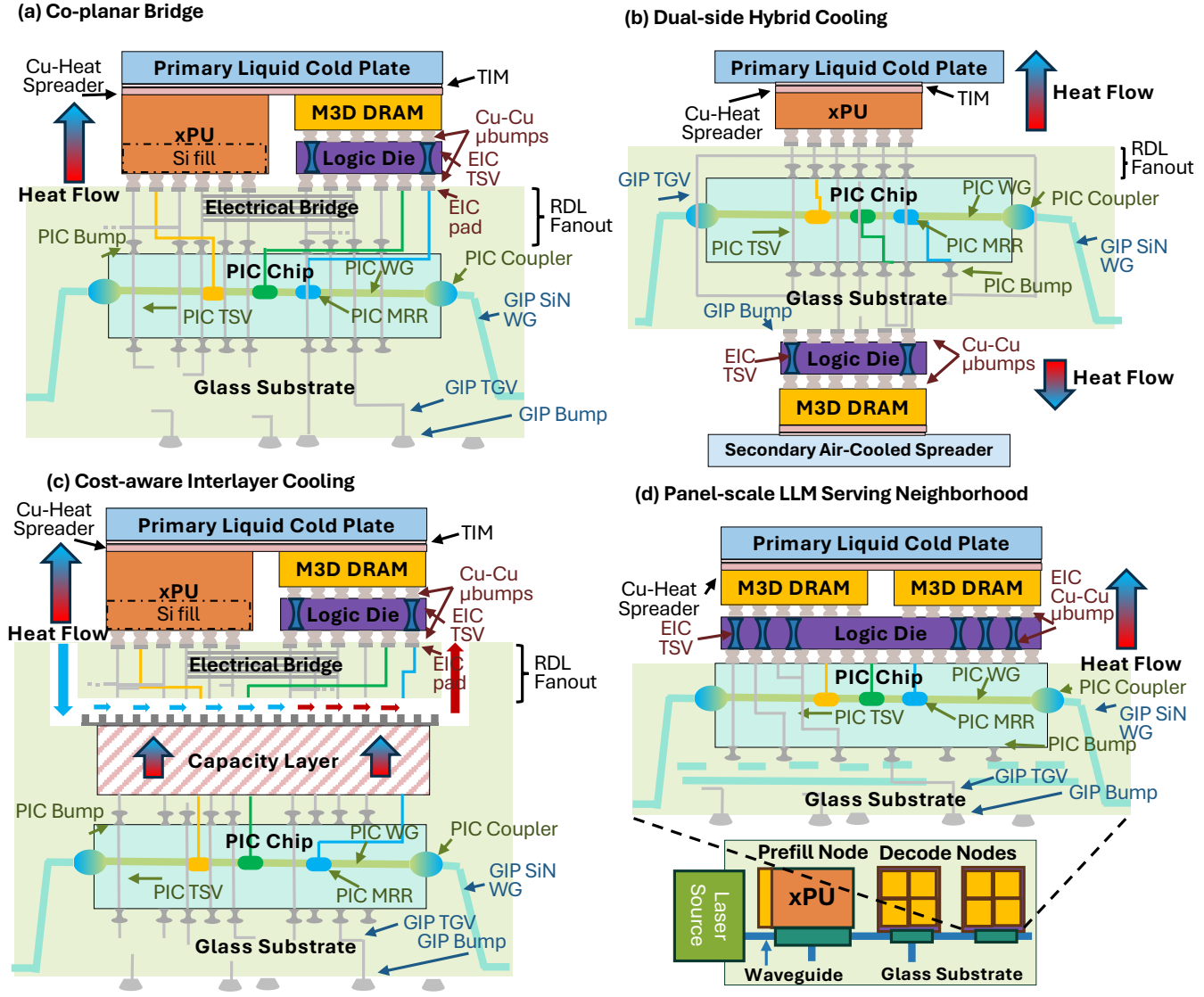

(a) Co-planar Bridge
Cu-Heat Spreader
Primary Liquid Cold Plate
TIM
xPU
Si fill
M3D DRAM
Logic Die
Cu-Cu µbumps
EIC TSV
Heat Flow
Electrical Bridge
EIC pad
RDL Fanout
PIC Bump
PIC Chip
PIC WG
PIC Coupler
PIC TSV
PIC MRR
GIP SiN WG
Glass Substrate
GIP TGV
GIP Bump
(b) Dual-side Hybrid Cooling
Primary Liquid Cold Plate
TIM
Heat Flow
Cu-Heat Spreader
xPU
RDL Fanout
GIP TGV
PIC Chip
PIC WG
PIC Coupler
PIC TSV
PIC MRR
GIP SiN WG
PIC Bump
Glass Substrate
GIP Bump
EIC TSV
Logic Die
Cu-Cu µbumps
Heat Flow
M3D DRAM
Secondary Air-Cooled Spreader
(c) Cost-aware Interlayer Cooling
Cu-Heat Spreader
Primary Liquid Cold Plate
TIM
xPU
Si fill
M3D DRAM
Logic Die
Cu-Cu µbumps
EIC TSV
Heat Flow
Electrical Bridge
EIC pad
RDL Fanout
Capacity Layer
PIC Bump
PIC Chip
PIC WG
PIC Coupler
PIC TSV
PIC MRR
GIP SiN WG
GIP TGV
GIP Bump
Glass Substrate
(d) Panel-scale LLM Serving Neighborhood
Primary Liquid Cold Plate
Cu-Heat Spreader
M3D DRAM
M3D DRAM
EIC Cu-Cu µbump
EIC TSV
Logic Die
Heat Flow
PIC Chip
PIC WG
PIC Coupler
PIC TSV
PIC MRR
GIP SiN WG
PIC Bump
GIP TGV
GIP Bump
Glass Substrate
Prefill Node
Decode Nodes
Laser Source
xPU
Waveguide
Glass Substrate